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Research Progress Report: Robust Parameter-Efficient Fine-Tuning for Multimodal Foundation Models under Structured Input Corruption

July 21, 2026

Abstract: This report summarizes ongoing exploratory work on robust parameter-efficient fine-tuning (PEFT) for multimodal document understanding models under structured input corruption (e.g., OCR noise). I describe the motivation for shifting away from classical noisy-*label* learning toward robustness against noisy *inputs*, present a prototype architecture (SALLoRA: Sensitivity-Aware Layer-wise LoRA with warm-up and sensitivity-aware masking), and report preliminary results on the FUNSD dataset. Results are derived from a single experimental configuration and are presented as early evidence to motivate further rigorous study, rather than as definitive conclusions.

1 Motivation

Document understanding models that jointly reason over text, layout, and visual features (e.g., LayoutLM-family architectures) are increasingly deployed on real-world scanned or photographed documents. In practice, the text stream fed to these models is rarely clean: it is the output of an OCR pipeline, and OCR errors (character substitutions, insertions, deletions, and segmentation mistakes) introduce *structured* corruption into the multimodal input.

This is a fundamentally different problem from the one addressed by the noisy-label learning literature. Classical noisy-label methods assume that the *input* is trustworthy and that only the *supervision signal* (the label) is corrupted. In document understanding under OCR noise, the situation is reversed: labels are typically clean (they come from human annotation on the underlying document), while the *input features* themselves are corrupted in a structured, position-dependent way that is correlated with document layout and token identity.

Parameter-efficient fine-tuning (PEFT) methods such as LoRA are attractive for adapting large pretrained backbones to this setting because they update a small number of parameters and are computationally efficient to retrain per deployment domain. However, standard LoRA fine-tuning lacks an explicit mechanism for distinguishing which parts of a noisy input are trustworthy, or for controlling how aggressively low-rank adapters should be updated when the training signal itself is degraded by corrupted inputs. This motivates the central question of this work:

> *"Can a PEFT framework be made explicitly robust to structured multimodal input corruption, by combining a conservative warm-up stage, a decoupled estimate of per-sample input quality,*

and a gradient-level sensitivity mask, rather than relying on the noisy-label learning toolbox alone?"

2 Contributions

In summary, this ongoing work makes the following core contributions:

- **Propose SALLoRA**, a sensitivity-aware PEFT framework specifically designed for structured multimodal corruption.
- **Introduce decoupled quality estimation and layer-wise masking** to improve model robustness and isolate clean signals during fine-tuning.
- **Present preliminary experiments** demonstrating improved resilience over standard LoRA on OCR-corrupted document understanding.

3 Literature Positioning

My review of the following three lines of work critically informed and refined the direction of this project.

3.1 Noisy-Label Learning

The broader literature on learning from noisy labels with deep neural networks provides a comprehensive taxonomy of approaches—loss correction, sample selection, robust loss functions, and label-noise-aware regularization—for the setting where labels, not inputs, are corrupted. While conceptually related (both settings involve training under corrupted supervision signals), the underlying assumption that inputs are clean does not transfer directly to the OCR-noise setting, as it is specifically the multimodal input representation that degrades.

3.2 CleaR

CleaR is representative of recent work that attempts to disentangle clean and corrupted signals at a finer granularity than sample-level reweighting, using representation-level cues to identify which parts of the training signal are reliable. This guided my design toward a similarly fine-grained (though input-side rather than label-side) notion of reliability: instead of asking "is this label trustworthy," I ask "is this token/region of the input trustworthy," which motivated the sensitivity-masking component of my prototype.

3.3 Robust and Generalized PEFT for Noisy Label Learning

Work on robust and generalized PEFT methods for noisy-label learning demonstrates that adapter-based fine-tuning can be made robust to label noise through careful decoupling of the adaptation process from unreliable supervision. This directly inspired my use of a *decoupled* quality estimator: a component that produces reliability weights without being differentiated through during the main task loss, aiming to prevent corrupted signals from directly degrading the adapter parameters via the estimator's own gradient path.

3.4 Resulting Shift in Direction

Taken together, these three lines of work motivated a shift in project framing: rather than treating OCR-corrupted document understanding as a noisy-label problem, I treat it as a noisy-*input* PEFT problem, and adapt the mechanisms above (loss-level robustness, fine-grained reliability estimation, and decoupled adapter training) to the input-corruption setting. I am not currently aware of a prior method that combines a warm-up-then-adapt curriculum, gradient sensitivity masking, and a decoupled quality estimator specifically for structured multimodal input corruption in document understanding; I present this combination strictly as an early-stage working prototype rather than a claim of state-of-the-art performance.

4 Method

I refer to the current prototype as **SALLoRA** (Sensitivity-Aware Layer-wise LoRA with robust warm-up and masked adaptation). It consists of six components layered on top of a frozen LayoutLMv3 backbone.

4.1 Architecture Components

Frozen backbone.

LayoutLMv3-base is kept entirely frozen; no backbone weights are updated at any stage, concentrating all adaptation capacity into a small number of additional parameters.

LoRA adapters.

A low-rank adapter ΔW_l (rank $r = 16$) is attached at every transformer layer l, injected on the hidden-state stream rather than exclusively inside attention projections.

Layer-wise learnable scaling.

Each layer has a scalar gate s_l, passed through a sigmoid $r_l = \sigma(s_l)$, that controls how much of that layer's LoRA update is mixed into the residual stream: $H_l = H_{l-1}^{\text{backbone}} + r_l \cdot \Delta W_l(H_{l-1})$. This is intended to allow the model to learn, per layer, how much to trust the adapter versus the frozen backbone representation.

Warm-up training.

Before any adapter is updated, only the classification head is trained on top of the frozen backbone's final hidden states. This aims to establish a reasonable decision boundary before any low-rank capacity is introduced, empirically anchoring later adaptation (see Section 6).

Gradient sensitivity masking.

After warm-up, a binary mask M_l is computed per adapter by taking a single gradient pass on a small clean subset of the training data and retaining only the top-$(100 - p)$% of LoRA parameters by gradient magnitude (I use $p = 80$, i.e., the top 20% most sensitive parameters). During adaptation, gradients are element-wise masked, $\Delta W_l.\text{grad} \leftarrow \Delta W_l.\text{grad} \odot M_l$, attempting to ensure that only the parameters identified as most responsive to clean signal are updated.

Decoupled quality estimator.

A small feed-forward network Q maps the mean-pooled embedding of a sample to a scalar quality weight $q_i \in [0.3, 0.7]$. Critically, q_i is computed under `torch.no_grad()` and utilized strictly as a fixed per-sample loss weight.

Quality-weighted optimization.

The per-sample token-classification loss ℓ_i is weighted by the corresponding detached quality score before averaging: $\mathcal{L} = \text{mean}_i(q_i \cdot \ell_i)$, down-weighting samples scored as lower quality by the estimator.

Note on the Quality Estimator: The current quality estimator is intentionally simplified. It serves primarily to validate the feasibility of the proposed training framework and is not yet a fully trained quality estimation model. Future work will replace this component with a fully supervised or meta-learned estimator.

4.2 Training Procedure

Training proceeds in two phases, outlined in Algorithm 1.

Computational Complexity:
SALLoRA introduces negligible inference overhead compared with standard LoRA because the additional computations consist only of lightweight layer-wise scaling and gradient masking during training. During inference, the model complexity is essentially identical to conventional LoRA.

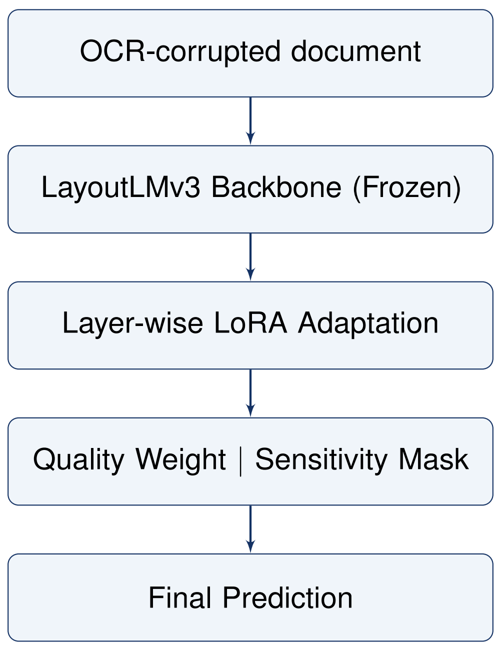

Figure 1: *SALLoRA Architecture Pipeline overview.*

5 Experimental Setup

- **Dataset:** FUNSD (Form Understanding in Noisy Scanned Documents), a document understanding benchmark with word-level bounding boxes and entity labels.
- **Backbone:** LayoutLMv3-base, kept fully frozen.
- **PEFT method:** LoRA, rank $r = 16$, applied at every transformer layer.
- **Training-time corruption:** Synthetic OCR-style corruption (character substitution, deletion, insertion, or a mixed `ocr_mix` scheme) is applied only to the training set, at a corruption level of 0.5.

Algorithm 1 SALLoRA: Robust Warm-up and Masked Adaptation

Require: Frozen backbone B with L layers; LoRA adapters $\{\Delta W_l\}$; layer scalars $\{s_l\}$; classifier head C; decoupled quality estimator Q; binary sensitivity mask $\{M_l\}$; training dataset D; warm-up epochs W; adaptation epochs N

Ensure: Trained $\{\Delta W_l\}$, $\{s_l\}$, C

Phase 1: Warm-up

Freeze B, $\{\Delta W_l\}$, $\{s_l\}$
for epoch $= 1$ to W **do**
  for each batch $(x, y) \in D$ **do**
    $H_L \leftarrow B(x)$
    $logits \leftarrow C(H_L)$
    $loss \leftarrow \text{CrossEntropy}(logits, y)$
    Update C only
  end for
end for

Phase 2: Sensitivity-aware adaptation

Unfreeze $\{\Delta W_l\}$, $\{s_l\}$
for epoch $= 1$ to N **do**
  for each batch $(x, y) \in D$ **do**
    $q_i \leftarrow Q(x)$ ▷ Decoupled quality weights
    $H_0 \leftarrow B_{\text{emb}}(x)$
    for $l = 1$ to L **do**
      $r_l \leftarrow \sigma(s_l)$
      $H_l \leftarrow H_{l-1}^{\text{backbone}} + r_l \cdot \Delta W_l(H_{l-1})$
    end for
    $logits \leftarrow C(H_L)$
    $ce_i \leftarrow (logits, y)$
    $\ell_i \leftarrow \text{mean}(ce_i \text{ over valid tokens})$
    $loss \leftarrow \text{mean}(q_i \cdot \ell_i)$ ▷ q_i treated as constant
    $loss.\text{backward}()$
    for $l = 1$ to L **do**
      $\Delta W_l.\text{grad} \leftarrow \Delta W_l.\text{grad} \odot M_l$ ▷ Sensitivity masking
    end for
    Update C, $\{s_l\}$, and masked $\{\Delta W_l\}$
  end for
end for
return $\{\Delta W_l\}, \{s_l\}, C$

- **Evaluation set:** The clean, uncorrupted FUNSD test set, so that reported F1 reflects generalization to clean documents despite noisy training.
- **Hyperparameters:** AdamW optimizer, weight decay 0.01, linear warm-up/decay LR schedule. Learning rate 2×10^{-4}. 2 warm-up epochs, 10 adaptation epochs.
- **Sensitivity mask:** Computed on a small clean subset (32 examples) via a single gradient pass, retaining the top 20% of LoRA parameters by gradient magnitude ($p = 80$).

6 Preliminary Single-Seed Results

Table 1 reports weighted F1 on the clean FUNSD test set for three configurations. Under 50% training-time corruption, unmodified LoRA shows a decrease of roughly 3.8 F1 points relative to the clean-data upper reference. Under the current experimental configuration, the SALLoRA prototype achieved a higher F1 than the clean-data baseline (0.7758 vs. 0.6710).

Table 1: *Weighted F1 on clean FUNSD test set.*

Method	Training Noise	F1 Score
Clean LoRA (upper reference)	0%	0.6710
Noisy LoRA (no robustness mechanisms)	50%	0.6334
Current Prototype (SALLoRA)	**50%**	**0.7758**

This unexpected observation requires further validation through repeated runs, stronger baseline comparisons, ablation studies, and statistical analysis. These preliminary results suggest potential regularization effects of the warm-up, masking, and weighting pipeline, rather than serving as a definitive demonstration that noisy training is broadly beneficial. **These results reflect a single experimental configuration and serve strictly as preliminary observations to guide future, more rigorous experimentation.**

7 Current Limitations & Planned Experiments

To set accurate, conservative expectations, I acknowledge several current limitations that form the basis for my immediate next steps:

1. **Statistical Reliability:** Results currently rely on a single random seed. *Next step: Repeat all configurations across multiple random seeds to establish variance estimates and run paired bootstrap significance tests.*
2. **Corruption Scope:** Only 50% corruption has been evaluated. *Next step: Sweep corruption levels (10% to 70%) and evaluate across isolated noise types (substitution vs. deletion vs. insertion).*
3. **Component Isolation:** The exact source of the performance gain is not yet decoupled. *Next step: Conduct rigorous ablation studies on warm-up, layer-wise scaling, quality weighting, and sensitivity masking individually.*
4. **Baselines:** The prototype has not been benchmarked against state-of-the-art robust methods. *Next step: Implement direct comparisons against CleaR and other baselines under matched training budgets.*
5. **Estimator & Mask Maturity:** The current estimator is fixed/simplified, and the sensitivity

mask relies on a static gradient pass. *Next step: Integrate a meta-learned quality estimator and evaluate an adaptive, per-batch dynamic masking variant.*

8 Conclusion

A thorough review of the noisy-label learning literature, CleaR, and recent robust PEFT methodologies critically shaped the trajectory of this research. Specifically, it motivated a pivot from treating OCR-corrupted document understanding as a traditional noisy-label challenge to framing it as a noisy-input PEFT problem.

While preliminary single-seed experiments on the FUNSD dataset suggest that the proposed architecture merits further investigation, these findings are strictly exploratory. The immediate next phase of this research is rigorous experimental validation. I look forward to discussing this framing, the prototype's viability, and the design of the planned experimental roadmap in the context of my prospective doctoral studies.